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Advances in IL-15–Based Cancer Immunotherapy and Divergent Immunological Effects of IL-2 and IL-15 Signaling via the Shared IL-2R $\beta\gamma$ Receptor

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Interleukin-15 (IL-15) has emerged as a central cytokine for next-generation cancer immunotherapy because of its unique ability to sustain the survival, proliferation, and cytotoxic function of memory CD8⁺ T cells and natural killer (NK) cells without promoting the expansion of regulatory T cells (Treg). These properties make IL-15 particularly attractive for achieving durable antitumor immunity, especially in solid tumors where immune persistence remains a major limitation. Although IL-15 shares the same signal-transducing receptor subunits (IL-2R β and the common γ chain) with interleukin-2 (IL-2), the two cytokines drive fundamentally different CD8⁺ T-cell fates, a distinction that underlies their markedly divergent therapeutic profiles in cancer immunotherapy. In recent years, multiple IL-15-based therapeutic strategies including recombinant IL-15, and IL-15 immunocytokines have entered clinical evaluation, demonstrating potent immune activation with manageable toxicity profiles. Recent clinical progress includes the FDA approval of Nogapendekin alfa inbakicept (N-803), the first IL-15-based immunotherapy approved for cancer treatment, alongside the advancement of other IL-15 superagonists into Phase II trials and growing evidence that IL-15 can enhance the efficacy of immune checkpoint blockade and engineered adoptive cell therapies such as CAR-T cells, CAR-NK cells, $\gamma\delta$ T cells, and invariant NKT cells. Despite these advances, important challenges remain, including cytokine-associated toxicities, optimal delivery strategies, and the immunosuppressive tumor microenvironment. This review summarizes recent progress in IL-15-based cancer immunotherapy, integrates emerging insights into IL-2R $\beta\gamma$ -driven CD8⁺ T-cell fate decisions, and discusses key opportunities and challenges for translating IL-15-mediated immune enhancement into durable clinical benefit.

KEYWORDS

cancer immunotherapy, CAR-NK cells, CAR-T cells, cytokine engineering, immune checkpoint inhibitor (ICI), immunocytokines, interleukin-15 (IL-15), interleukin-2 (IL-2)

1 Introduction

Immunotherapy for cancer has fundamentally revolutionized modern oncological strategies. Compared to conventional chemotherapy and radiation, immunotherapy harnesses the immune system's intrinsic capacity to recognize and eliminate malignant cells, thereby offering a distinct and often more durable therapeutic approach (1). This paradigm shift has driven the development of a broad spectrum of cancer immunotherapeutic agents, including recombinant immunostimulatory cytokines (e.g., interleukin-2 (IL-2) and interferon IFN- α), immune checkpoint inhibitors targeting PD-1/PD-L1, and adoptive cell therapies such as chimeric antigen receptor (CAR)-T cells (2–4). Despite these remarkable advances, significant challenges remain, such as the identification of optimal immunological targets, mitigation of therapy-associated toxicities, and reliable prediction of patient responsiveness. These challenges have prompted extensive investigation into strategies that enhance efficacy while minimizing toxicity in cancer immunotherapy.

Cytokines represent a critical class of immunomodulatory molecules capable of shaping both innate and adaptive immune responses by acting on effector and cytotoxic immune cells. Extensive preclinical studies across diverse animal tumor models have demonstrated potent antitumor activity for multiple cytokines, including IL-2, granulocyte-macrophage colony-stimulating factor (GM-CSF), IL-12, IL-15, IL-18, IL-7, and IL-21. (5, 6). Consequently, IL-2 was the first immunotherapy approved by the United States Food and Drug Administration (FDA) for cancer treatment in 1990s (7). Encouraged by the efficacy of IL-2, a variety of cytokines including granulocyte-macrophage colony-stimulating factor (GM-CSF) and several interleukins such as IL-15, IL-12, IL-18, IL-7, and IL-21, have been evaluated in clinical trials involving patients with advanced malignancies (8). However, despite its early clinical success, the clinical use of IL-2 has been declined largely due to its severe systemic toxicities, most notably vascular leak syndrome (VLS) (9). IL-2 and IL-15 are closely related members of the common γ -chain cytokine family and play central roles in regulating T-cell immunity. Both cytokines signal through a shared receptor complex composed of IL-2R β (CD122) and the common γ chain (CD132), activating overlapping intracellular pathways that control T-cell proliferation, survival, and differentiation. IL-15, originally identified based on its ability to support T-cell growth, was subsequently recognized as a critical regulator of memory CD8⁺ T cells and NK cells, positioning it as a promising alternative to IL-2 for therapeutic immune activation. Despite their shared receptor usage and signaling machinery, IL-2 and IL-15 exert strikingly different effects on T-cell fate and immune homeostasis, a distinction that has profound implications for cancer immunotherapy (10). Compared to IL-15, IL-2 is able to maintain regulatory T cells (Tregs), which can suppress antitumor immune responses (11). Moreover, IL-2 promotes activation-induced cell death (AICD) of CD8⁺ effector T cells, thereby limiting sustained antitumor immunity (8). Consistent with these functional differences, genetic deficiency of IL-2 leads to immune dysregulation and severe autoimmunity (12–15), whereas IL-15 deficiency results in an almost complete absence of

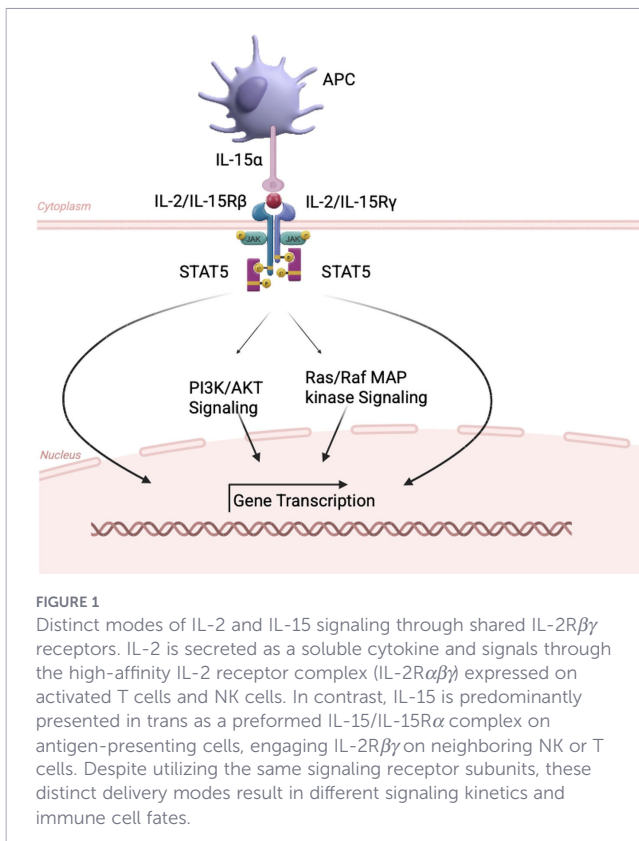
NK cells and a profound reduction in effector and memory CD8⁺ T-cell populations (16).

Importantly, preclinical studies have further revealed distinct toxicity profiles between the two cytokines. IL-15 administration is associated with minimal vascular capillary leak syndrome compared to IL-2, suggesting a more favorable therapeutic window (17). Collectively, these biological and functional distinctions support the premise that IL-15 may represent a more efficacious and better-tolerated cytokine for cancer immunotherapy. In this review, we summarize the biology and immunological functions of IL-15, highlight the mechanistic divergence between IL-2- and IL-15-mediated signaling, examine the current landscape of IL-15-based clinical strategies, and discuss future directions for optimizing cytokine-supported immunotherapy in solid malignancies.

2 The biology of IL-15 and IL-15 receptors

IL-15 was independently identified in 1994 by two research groups based on the observation that culture supernatants derived from CV-1/EBNA simian kidney epithelial cells or the human T-cell leukemia virus-1 (HTLV-1)-transformed T-cell line HuT-102 supported proliferation of the IL-2-dependent CTLL-2 T-cell line despite neutralization of IL-2, indicating the presence of a distinct T-cell growth factor. The activity present in HuT-102 supernatants was purified and initially termed IL-T (18, 19). In parallel, the corresponding factor was isolated from CV-1/EBNA cell culture supernatants and designated IL-15 (20). IL-15 is a 14–15 kDa glycoprotein encoded on human chromosome 4q31 and is expressed across a wide range of cell types. (21). Structurally, IL-15 is a class I helical cytokine with a four- α -helix bundle fold in which the helices are arranged in the characteristic up-up-down-down topology and is a member of the class I four-helix bundle cytokine family, which includes γ c-dependent cytokines such as IL-2, IL-4, IL-7, and IL-9 (22). Physiologically, IL-15 is co-expressed with its high-affinity receptor IL-15R α and is produced in limited amounts as a soluble cytokine. Instead, IL-15 is predominantly delivered through trans-presentation by antigen-presenting cells (APCs), in which IL-15 is bound to IL-15R α on the surface of dendritic cells (DCs) and macrophages and presented to neighboring lymphocytes expressing the IL-2/IL-15R β and common γ chains (23). In this process, IL-15R α functions as a chaperone that stabilizes IL-15 intracellularly and traffics it to the plasma membrane for focused delivery to NK cells, NKT cells, and memory CD8⁺ T cells that express the shared IL-2R $\beta\gamma$ signaling complex (24, 25) (Figure 1).

At the genetic level, IL-15 expression is regulated by a uniquely complex gene architecture that limits uncontrolled cytokine release. Although IL-15 mRNA is constitutively expressed across many cell types, including DCs, monocytes, macrophages, bone marrow stromal cells, and intestinal epithelial cells (26, 27), translation of IL-15 is tightly constrained by multiple regulatory elements, including upstream AUGs, alternative signal peptides, and inefficient secretion motifs. As a result, IL-15 protein is



predominantly produced and displayed by monocytes, macrophages, and DCs, rather than being freely secreted. This multilayered post-transcriptional and intracellular control ensures that IL-15 activity is spatially restricted and delivered primarily through receptor-mediated trans-presentation rather than systemic diffusion.

IL-15 is tightly linked to innate immune activation and is upregulated in response to inflammatory and microbial stimuli. In APC, IL-15 expression increases following exposure to bacterial components such as lipopolysaccharide (LPS), type I IFN- α/β , type II IFN- γ , double-stranded RNA, and viral infection (28, 29). These signals promote coordinated induction of IL-15 and IL-15R α , enabling receptor-mediated trans-presentation to neighboring lymphocytes. This mode of IL-15 delivery is essential for the generation, maintenance, and functional specialization of IL-15-responsive immune populations, particularly NK cells and memory CD8⁺ T cells (23, 30, 31).

3 Divergent immunological effects of IL-2 and IL-15 signaling through the shared IL-2Rβγ receptor

As we mentioned before, IL-2 and IL-15 are members of the common γ -chain (γ c) cytokine family, together with IL-4, IL-7, IL-9, and IL-21. Both cytokines transmit signals through a

receptor complex consisting of IL-2 receptor β (IL-2R β , CD122) and the common γ -chain (CD132), thereby activating overlapping intracellular signaling cascades, including the JAK1/JAK3-STAT5, PI3K-AKT, and MAPK pathways. Early studies therefore proposed that IL-2 and IL-15 might exert largely overlapping biological functions. The distinct genomic, cellular, and functional characteristics of these two cytokines that differentiate their biological roles are summarized in Table 1. Subsequent clinical and experimental observations, however, have clearly demonstrated that engagement of this shared receptor complex can result in fundamentally different immune and therapeutic outcomes. These differences are particularly evident in clinical oncology. High-dose IL-2 therapy is capable of inducing durable tumor regression in a subset of patients with melanoma and renal cell carcinoma, but its broader application is constrained by severe systemic toxicity, preferential expansion of regulatory T cells, and induction of activation-induced cell death and functional exhaustion in effector T cells. In contrast, IL-15 preferentially sustains the survival, proliferation, and long-term maintenance of NK cells and memory CD8⁺ T cells, without driving regulatory T-cell expansion or terminal effector differentiation. These properties make IL-15 a more favorable cytokine for therapeutic strategies aimed at promoting durable antitumor immunity. The divergent biological effects of IL-2 and IL-15 do not arise from differences in their core IL-2Rβγ signaling machinery, but instead reflect variations in receptor distribution, cytokine presentation, and context-dependent signal integration. A major determinant of this divergence is the differential utilization of cytokine-specific α -receptor chains. IL-2 binds with high affinity to IL-2R α (CD25), which is transiently expressed on activated T and B cells and constitutively expressed on regulatory T cells. In contrast, IL-15 associates with IL-15R α , a receptor that is predominantly expressed by activated monocytes and dendritic cells (9, 32).

Beyond differences in receptor expression, IL-2 and IL-15 also diverge in downstream signal integration. Although both cytokines activate the JAK1/JAK3-STAT5 axis, IL-15-induced T-cell proliferation relies strongly on FcBP12-mediated activation of p70 S6 kinase (p70S6K), whereas FcBP12 is dispensable for IL-2-driven proliferation (33). Consistent with this mechanism, FcBP12 deficiency selectively impairs ERK and p70S6K phosphorylation in response to IL-15 but not IL-2, indicating that IL-15 engages a distinct proliferative signaling module (33). Conversely, FcBP12.6 contributes to IL-2 signaling but not to IL-15, further reinforcing their mechanistic divergence (33). The most critical distinction between IL-2 and IL-15, however, lies in their modes of cytokine presentation. IL-2 is primarily secreted as a soluble cytokine or retained within the extracellular matrix, where it engages pre-assembled high-affinity IL-2Rαβγ complexes on activated lymphocytes (24). This signaling configuration preferentially targets CD25-expressing cells, including regulatory T cells, thereby promoting immune regulation and terminal effector differentiation. In contrast, IL-15 is rarely released in a free form. Instead, it is co-expressed intracellularly with IL-15R α and retained

on the surface of monocytes and dendritic cells. Inflammatory stimulation by type I or type II interferons, together with NF- κ B activation through CD40 or Toll-like receptor pathways, induces coordinated expression of IL-15 and IL-15R α , resulting in the formation of stable membrane-bound complexes that recycle through endosomal compartments and persist on the cell surface (24).

Within immunological synapses, IL-15 is delivered in trans by IL-15R α on antigen-presenting cells to neighboring lymphocytes that express IL-2R $\beta\gamma$ but lack IL-15R α (34, 35). The principal targets of this trans-presentation mechanism are NK cells and memory CD8⁺ T cells, which express high levels of IL-2R β but minimal CD25. Although CD25 on dendritic cells can transiently present IL-2 in trans to antigen-specific T cells during early immune responses (24), this interaction does not reproduce the sustained, spatially restricted signaling characteristic of IL-15. In addition to trans-presentation, IL-15 can signal in cis on cells that co-express IL-15R α , IL-2R β , and the common γ -chain. This mode of signaling is particularly important for optimal NK-cell activation during inflammatory responses, as demonstrated in lipopolysaccharide-driven models (34). Moreover, IL-15 can engage non-canonical signaling pathways in certain innate immune cell types. In mast cells, IL-15 signals through alternative receptor complexes that recruit JAK2 and STAT5 rather than the classical JAK1/JAK3 pathways used in T cells (35). In myeloid cells, IL-15 can also transduce signals independently of IL-2R $\beta\gamma$ and JAK/STAT through IL-15R α , JNK, and NF- κ B pathways to induce chemokine production, including RANTES (34).

Together, these receptor-level and signaling-context distinctions explain how IL-2 and IL-15, despite sharing IL-2R $\beta\gamma$, drive fundamentally different CD8⁺ T-cell fates. IL-2 signaling preferentially enforces regulatory and terminal effector programs, whereas IL-15 supports the persistence of long-lived, cytotoxic, and memory lymphocyte populations. IL-15-based cancer immunotherapies exploit this biology by delivering IL-15 through cis- or trans-presentation mechanisms that selectively engage IL-2R $\beta\gamma$ on tumor-reactive lymphocytes, thereby promoting durable antitumor immunity while avoiding the immunosuppressive liabilities associated with IL-2.

Although IL-15 is best known for its critical role in the maintenance and activation of NK cells and memory CD8⁺ T cells, accumulating evidence indicates that IL-15 also influences several other immune cell populations relevant to tumor immunity. IL-15 can enhance the survival and functional activity of CD4⁺ T cells, particularly promoting the expansion of effector and memory subsets under inflammatory conditions. In addition, IL-15 has been shown to modulate dendritic cells, macrophages, and innate lymphoid cells, contributing to enhanced antigen presentation, cytokine production, and coordination of innate-adaptive immune responses. IL-15 signaling can also influence $\gamma\delta$ T cells and NKT cells, which possess innate-like cytotoxic properties and are increasingly explored as platforms for cancer immunotherapy. Collectively, these findings highlight the broader immunomodulatory role of IL-15 across multiple immune compartments and further support its therapeutic potential in cancer immunotherapy.

4 IL-15 based cancer immunotherapy

4.1 rhIL-15

The clinical development of IL-15-based agents has progressed significantly, with several candidates currently being evaluated in human trials (Table 2). Central to these advancements, the clinical manufacture of recombinant human interleukin-15 (rhIL-15) has been established using *Escherichia coli* (*E. coli*), providing a robust foundation for experimental and translational studies of IL-15-based immunotherapy (42). Owing to improvements in rhIL-15 purification and formulation, administration of rhIL-15 has been shown to induce tumor regression and suppress metastatic dissemination in murine cancer models, including the LA795 lung adenocarcinoma model. Tang and colleagues (43) demonstrated that rhIL-15 treatment resulted in a significantly lower tumor burden and prolonged survival compared with rhIL-2 in the LA795 transplantable lung adenocarcinoma model, indicating a superior anti-tumor efficacy of rhIL-15 at equivalent doses. In addition, IL-15-mediated anti-tumor activity has been widely reported across other transplantable tumor systems, including melanoma, colon carcinoma, carcinoma-derived lines, and lymphomas in preclinical studies (44).

In addition to its therapeutic efficacy, the safety of rhIL-15 has been evaluated in multiple preclinical and clinical studies. Preclinical investigations demonstrated that rhIL-15 exhibits an acceptable safety profile across several dosing regimens. Administration of rhIL-15 in non-human primates by continuous intravenous infusion at 20 μ g/kg/day for 10 days was well tolerated, with no lasting effects on clinical status, body weight, or food consumption. At higher doses (40 μ g/kg/day), treatment resulted in dose-dependent but transient laboratory abnormalities, including reversible elevations in liver enzymes and occasional neutropenia, all of which resolved following treatment discontinuation. Importantly, no progressive or irreversible toxicities were observed, supporting the overall tolerability of short-term IL-15 administration (45). In this study, IL-15 administration also induced marked expansion of circulating NK cells and effector memory CD8⁺ T cells, highlighting its potential to promote durable anti-tumor immunity (45). Consistent with its physiological role in lymphocyte homeostasis, IL-15 preferentially expands NK cells and effector-memory CD8⁺ T cells *in vivo*, as demonstrated in rhesus macaques (46). Building on these encouraging preclinical findings, a phase I dose-escalation trial evaluated subcutaneous rhIL-15 in patients with advanced refractory solid tumors, including melanoma, renal cell carcinoma, non-small cell lung cancer, and head and neck cancer (47) (ClinicalTrials.gov identifier: NCT01727076). In this clinical trial, rhIL-15 was administered as daily injections for two weeks of a four-week cycle, with dose levels ranging from 0.25 to 3.0 μ g/kg/day. The primary objectives were to assess safety, tolerability, and the maximum tolerated dose, as well as pharmacodynamic immune responses and preliminary clinical activity. rhIL-15 was generally well tolerated; however, dose-limiting toxicities at higher dose levels included pancreatitis and transient cardiovascular effects, establishing 2 μ g/kg/day as the maximum tolerated dose. Although no objective tumor

regressions were observed, several patients achieved disease stabilization, supporting continued development of IL-15-based therapies, particularly in combination with other immunotherapeutic strategies.

Despite its promising biological activity, rhIL-15 monotherapy faces several important limitations. Effective expansion and maintenance of immune effector cells require sustained exposure to IL-15 at biologically active concentrations (48). However, maintaining durable levels of soluble IL-15 (sIL-15) *in vivo* is challenging due to its short serum half-life, reported to be approximately 2–3 hours following intravenous administration, together with relatively rapid systemic clearance (5–11 L/h depending on dose) (47, 48). A key determinant of IL-15 stability and bioavailability is its high-affinity receptor IL-15R α , which governs IL-15 trafficking, presentation, and persistence in circulation (23). Collectively, these constraints limit the therapeutic efficacy of unmodified IL-15 and have driven the development of engineered IL-15 derivatives designed to improve pharmacokinetics, receptor engagement, and targeted delivery.

4.2 IL-15 modification and IL-15/IL-15R α complex

To address the limitations associated with rhIL-15, two major optimization strategies have been explored. The first approach involves engineering IL-15 itself through mutations or other

molecular modifications, a strategy commonly used to improve the pharmacological properties of protein therapeutics. The second strategy focuses on the development of IL-15/IL-15R α complexes. Because IL-15 naturally functions as a heterodimeric IL-15-IL-15R α complex *in vivo* (hetIL-15), researchers hypothesized that mimicking this physiological form could enhance cytokine stability and biological activity compared with IL-15 monomers. Subsequent studies have supported this concept, demonstrating improved pharmacokinetic and immunostimulatory properties for IL-15/IL-15R α -based agonists. Notably, nogapendekin alfa inbakicept (N-803) is currently the first and only IL-15-based immunotherapy approved by the U.S. FDA, representing a major milestone in the clinical translation of IL-15 agonists.

N-803 was named after being purchased by ImmunoBio after the name of ALT-803 which is a 2 (IL-15/IL-15R α) tetramer formed by the assembly of two IL-15 mutants (IL-15 N72D, which has a higher affinity for the receptor) and two IL15R α Su-IgG1 Fc fusion proteins (49). The combination of N-803 with *Bacillus Calmette-Guérin* (BCG) was approved by the FDA for patients with BCG-unresponsive non-muscle invasive bladder cancer (NMIBC) (36). This approval was supported by the results of the multicenter trial (NCT0302285), which evaluated intravesical N-803 together with BCG in patients with high-risk NMIBC with carcinoma *in situ* with or without papillary tumors (50). In this study, 62% of the 77 enrolled patients achieved a complete response, and 58% of these responses were sustained for at least 12 months, exceeding the benchmark complete response rate of 30% at 1 year proposed by FDA experts and the International Bladder Cancer Group (51–53). The development of this therapeutic strategy was motivated by earlier preclinical studies aiming to enhance BCG-mediated antitumor immunity. Although intravesical BCG can induce durable responses in NMIBC, approximately 40% of patients experience disease recurrence and become unresponsive to further BCG therapy (54–56). Preclinical bladder cancer models demonstrated that the combination of IL-15 and intravesical BCG administration was safe and significantly enhanced antitumor immune responses (57–59). While the precise mechanisms were not fully elucidated, improved outcomes were associated with the expansion of NK cells and increased production of inflammatory cytokines. Collectively, these findings provided the rationale for clinical evaluation of the N-803 and BCG combination in NMIBC patients.

In addition, FIST15 which links the IL-15R α sushi domain-IL-15 complex to tandem TGF- β traps derived from TGF- β RII-ECD, induces potent NK cell-dependent antitumor activity in syngeneic mouse models (63). Building on this concept, HCW9218 was generated using a soluble tissue factor-based scaffold to integrate the IL-15R α -IL-15 complex with TGF- β RII-ECD. This bifunctional molecule promoted intratumoral accumulation of NK cells and CD8 $^{+}$ T cells and induced strong antitumor responses in preclinical models (60). In early clinical testing, HCW9218 was well tolerated and induced marked NK-cell expansion following a single dose, with dose escalation ongoing up to 0.5 mg/kg (NCT05322408). Beyond IL-15 mutations and IL-15/IL-15R α complex engineering, another important strategy to enhance the therapeutic potential of IL-15 involves the development of IL-15-based immunocytokines.

TABLE 1 Comparative features of IL-2 and IL-15 biology and receptor engagement.

Property	IL-2	IL-15
Genomic organization	Composed of four exons and located on chromosome 4q26	Encoded by eight exons and mapped to chromosome 4q31
Primary cellular sources	Predominantly produced by activated CD4 $^{+}$ Th1 cells	Mainly synthesized by activated DCs and monocytes
Regulation of expression	Controlled at the transcriptional level and through mRNA stabilization	Largely regulated post-transcriptionally, including translational control and intracellular trafficking
Mode of receptor engagement	Signals primarily via cis-interaction with IL-2R α , IL-2/IL-15R β , and the common γ -chain co-expressed on activated T and B cells	Presented as an IL-15/IL-15R α complex on antigen-presenting cells and delivered in trans to NK cells and CD8 $^{+}$ memory T cells expressing IL-2/IL-15R β and the common γ -chain
Principal immunological role	Supports regulatory T-cell homeostasis and enforces peripheral tolerance through activation-induced cell death of autoreactive lymphocytes	Sustains NK cells and CD8 $^{+}$ CD44 hi memory T cells, enabling durable protective immune responses
Phenotype of cytokine or private receptor α -chain deficiency in mice	Severe lymphoproliferation with enlarged peripheral lymphoid organs and autoimmune manifestations due to uncontrolled T- and B-cell expansion	Profound loss of NK, NKT, $\gamma\delta$ T cells, and CD8 $^{+}$ CD44 hi memory T-cell populations

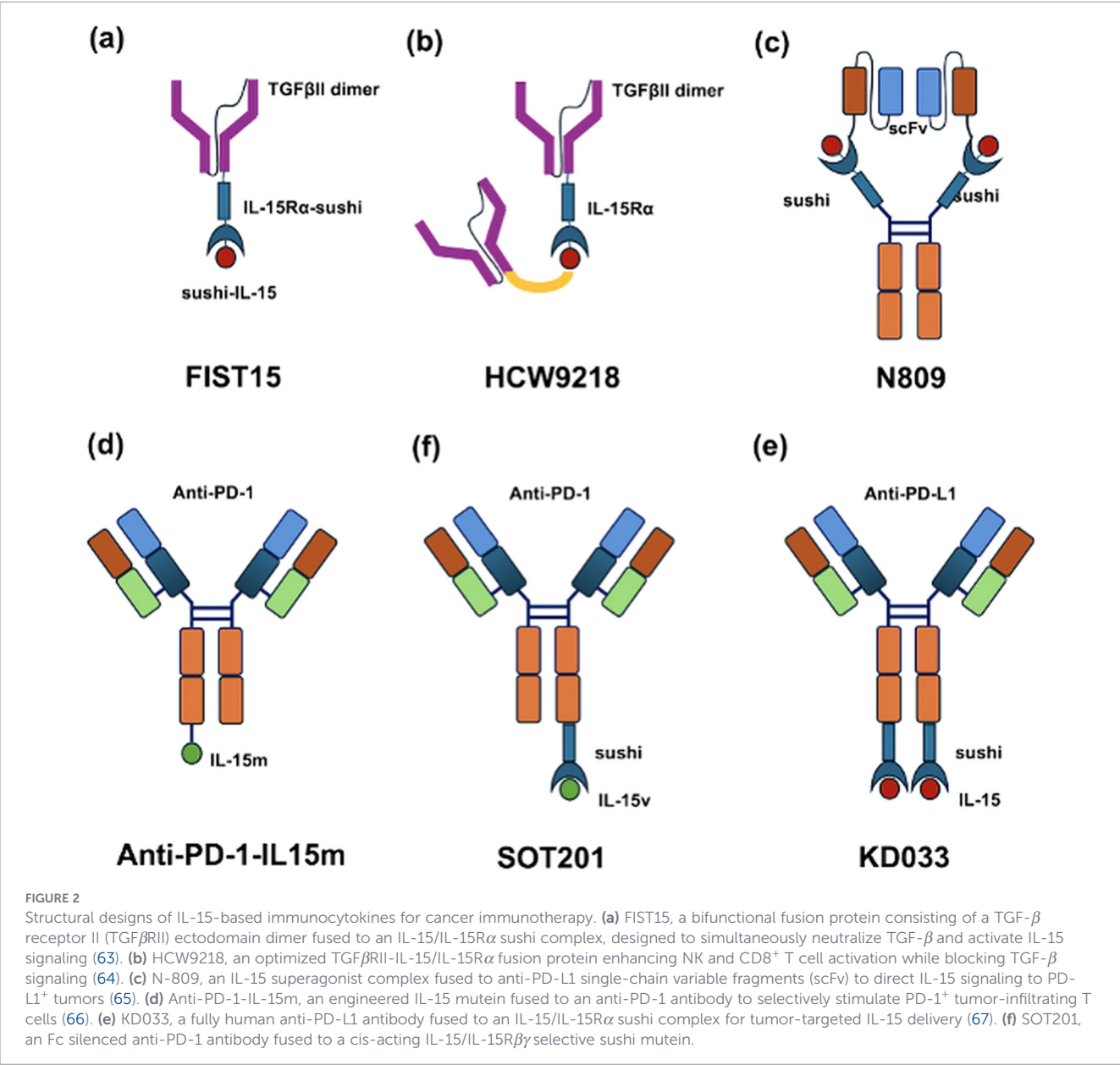
TABLE 2 Recent clinical advances in IL-15–based cancer immunotherapy.

Agent/strategy	Indication	Clinical stage	Key findings	Ref.
Anktiva (N-803)	NMIBC	FDA Approved	High complete response; durable remissions	(36)
N-803 post-HCT	AML/MDS	Phase I/II	NK expansion; antitumor surveillance	(37)
NKTR-255 + CAR-T	B-ALL/LBCL	Phase I	Enhanced immune expansion; early efficacy	(38)
NIZ985 (IL-15/IL-15Rα)	Solid tumors	Phase I	Cytokine induction; lymphocyte proliferation	(39)
SOT201 (PD-1–IL-15 mutein)	Solid tumors	Phase I	Selective stimulation of PD-1 ⁺ TILs	(40)
PF-07209960 (anti-PD-1–IL-15m)	Solid tumors	Phase I	PK/PD and safety reported	(41)

4.3 IL-15-based immunocytokines

Antibody-cytokine fusion proteins, commonly referred to as immunocytokines, represent a targeted strategy for cytokine delivery and have been extensively developed and evaluated in clinical studies across a broad range of oncologic and inflammatory indications. These engineered molecules typically comprise a targeting antibody domain linked via a flexible peptide linker to a

cytokine effector. By concentrating cytokine activity at disease sites, immunocytokines can reduce systemic exposure and toxicity associated with conventional cytokine therapy, thereby expanding the therapeutic window of the cytokine payload (61, 62). To enhance and spatially restrict IL-15-mediated immune activation, multiple IL-15-based immunocytokines have been engineered, as illustrated in Figure 2, including fusion constructs targeting immune checkpoint molecules such as PD-1 and PD-L1, as well



as immunomodulatory proteins such as the transforming growth factor- β receptor II ectodomain (TGF- β RII-ECD). Several of these molecules are currently in preclinical and early-stage clinical development and have demonstrated robust immune-stimulating activity in murine tumor models, primarily through expansion and functional enhancement of NK cells and CD8⁺ T cells.

In addition to enabling tumor-targeted cytokine delivery, antibody-based immunocytokines substantially improve the pharmacokinetic properties of IL-15. In contrast to soluble rhIL-15, IL-15/IL-15R α complexes and Fc-containing immunocytokines demonstrate markedly prolonged systemic persistence due to receptor stabilization and FcRn-mediated recycling. Preclinical and clinical studies have reported circulatory half-lives ranging from approximately 20–30 hours for IL-15/IL-15R α superagonist complexes to several days for antibody-based IL-15 immunocytokines, resulting in improved bioavailability and sustained activation of NK cells and CD8⁺ T cells. These pharmacokinetic improvements provide a key rationale for the development of targeted IL-15 immunocytokine platforms.

Several next-generation IL-15 immunocytokines have further integrated cytokine signaling with immune checkpoint blockade to enhance tumor-selective immune activation. N-809 is a bifunctional fusion protein composed of an IL-15/IL-15R α superagonist linked to an IgG1 Fc domain bearing two anti-PD-L1 single-chain variable fragments. In murine models of breast and colon carcinoma, N-809 demonstrated superior antitumor activity compared with the combination of non-targeted IL-15 and anti-PD-L1. This efficacy was associated with enhanced activation and recruitment of NK cells and CD8⁺ T cells within tumor-draining lymph nodes and the tumor microenvironment, as well as chemokine-mediated effector cell trafficking into tumors (64, 68). An alternative approach was developed by Pfizer with anti-PD-1-IL-15m, an immunocytokine in which a PD-1-specific antibody is fused to an IL-15 mutein lacking IL-15R α binding and exhibiting reduced affinity for IL-2R $\beta\gamma$. This design reduced receptor-mediated clearance and prolonged systemic exposure relative to wild-type anti-PD-1-IL-15, resulting in improved pharmacokinetic persistence and sustained immune activation *in vivo*. In preclinical studies, a single subcutaneous dose of anti-PD-1-IL-15m elicited strong, dose-dependent antitumor responses that exceeded those achieved with combined IL-15 superagonist and anti-PD-1 therapy. However, dosing at 5 mg/kg was associated with body-weight loss driven by peripheral NK-cell expansion. Mechanistically, tumor control depended primarily on CD8⁺ T cells rather than NK cells, with selective expansion of a proliferative, effector-like exhausted CD8⁺ T-cell population within tumors (66). Another PD-1-targeted immunocytokine, SOT201, consists of an Fc-silenced humanized anti-PD-1 antibody fused to a covalent sushi/IL-15 mutein with reduced IL-2R $\beta\gamma$ affinity. This construct selectively stimulates PD-1⁺ tumor-infiltrating lymphocytes, restoring exhausted T-cell function *in vitro* and expanding antigen-specific PD-1⁺ CD8⁺ T cells *in vivo*. SOT201 demonstrated potent antitumor activity in both PD-1-sensitive and PD-1-resistant tumor models and exhibited favorable pharmacokinetics with prolonged systemic persistence in non-human primates, supporting its ongoing evaluation in a phase I clinical trial (NCT06163391) (69).

In addition to PD-1-targeted designs, KD033 is a homodimeric immunocytokine composed of a fully human anti-PD-L1 antibody fused to the IL-15R α sushi domain-IL-15 complex. In syngeneic tumor models, a single dose of KD033 induced durable antitumor responses with a favorable safety profile, reflecting the extended systemic persistence and improved bioavailability conferred by the IL-15/IL-15R α complex and antibody scaffold. (67). Early clinical evaluation further demonstrated that KD033 was generally well tolerated, with manageable adverse events.

4.4 IL-15 in adoptive cell therapy

Genetic modification has become central to the development of adoptive cell therapies for cancer, enabling precise control over immune cell survival, function, and persistence. Current engineering approaches include the introduction of cytokine or chemokine support, modulation of transcriptional programs, disruption of inhibitory immune checkpoints, and incorporation of inducible safety switches (70). Among these strategies, the ectopic expression of human IL-15 has emerged as a particularly powerful method to enhance the therapeutic durability and antitumor activity of engineered immune cells, as summarized in the chronological overview of clinical trials involving IL-15-armored cellular therapies (Table 3).

In contrast to soluble cytokine therapies, the persistence of IL-15-engineered cellular therapies is not primarily determined by cytokine pharmacokinetics but rather by the *in vivo* expansion and survival of the engineered immune cells. Clinical and preclinical studies have shown that CAR-T and CAR-NK cells expressing IL-15 can persist in circulation for several weeks to months, reflecting sustained autocrine or membrane-associated IL-15 signaling that promotes long-term cell survival and functional competence (71). This prolonged cellular persistence represents a key advantage of IL-15-armored adoptive cell therapies compared with systemic cytokine administration.

IL-15 has been incorporated into a wide range of CAR-based cellular platforms, including conventional T cells (71–73), natural killer (NK) cells (74, 75), invariant NKT cells (76–79), $\gamma\delta$ T cells (80), and macrophages (81). Across these diverse immune lineages, IL-15 expression consistently promotes improved cell survival, sustained proliferation, and enhanced cytotoxic function *in vivo*, thereby addressing key limitations of conventional adoptive cell therapies.

4.4.1 IL-15-armored CAR T cells

To enhance the persistence, expansion, and antitumor efficacy of CAR-T cells, multiple groups have explored genetic strategies to incorporate IL-15 signaling directly into CAR-T cell products. Early preclinical studies demonstrated that constitutive or regulated IL-15 expression could overcome key limitations of conventional CAR-T cells, including activation-induced cell death, functional exhaustion, and insufficient *in vivo* persistence. Hoyos and colleagues first reported a CD19-targeted CAR-T construct co-expressing IL-15 together with an inducible caspase-9 (iC9) suicide gene (iC9/CAR.19/IL-15), which significantly enhanced CAR-T expansion,

TABLE 3 Chronologically ordered clinical trials of IL-15-armed cellular therapies in cancer immunotherapy.

Start year	Therapy	Malignancies	Description (cell type, target, IL-15 design)	Clinical trial
2015	CD19 CAR-T (IL-15 armored)	Recurrent DLBCL; recurrent FL; recurrent MCL	Autologous CD19-targeted CAR-T cells engineered to express IL-15 to enhance T cell persistence and antitumor activity; IL-15 construct details not publicly disclosed	NCT02652910
2017	CD19 CAR-T (IL-15 armored)	B-lymphoid malignancies; ALL; CLL; NHL	Autologous CD19-directed CAR-T cells with IL-15 co-expression to promote T-cell expansion and survival; IL-15 expression cassette reported but promoter and variant not specified	NCT03056339
2019	CD19 CAR-T (IL-15 armored)	MCL; relapsed/refractory DLBCL; FL; B-cell NHL	Autologous CD19 CAR-T cells engineered with IL-15 signaling support to enhance proliferation and persistence; IL-15 construct design not disclosed	NCT03579927
2019	GD2 CAR-T (IL-15 armored)	Neuroblastoma; osteosarcoma	GD2-targeting CAR-T cells engineered to express IL-15 to improve persistence and antitumor immune activity; IL-15 variant and promoter not reported	NCT03721068
2020	CD19 CAR-T (IL-15 armored)	Refractory B-NHL; SLL; relapsed ALL; relapsed CLL; relapsed NHL	CD19-directed CAR-T cells with IL-15 co-expression to enhance immune expansion and persistence; IL-15 expression cassette details not publicly available	NCT03774654
2020	GPC3 CAR-T (IL-15 armored)	Hepatocellular carcinoma; hepatoblastoma	GPC3-targeting CAR-T cells engineered to express IL-15 to improve antitumor activity in solid tumors; IL-15 construct details not reported	NCT04093648
2021	CD19 CAR-T (IL-15 armored)	ALL; B-cell lymphoma; CLL	CD19-targeted CAR-T cells engineered with IL-15 expression to support proliferation and survival of transferred T cells; IL-15 promoter and variant not specified	NCT04814004
2021	GPC3 CAR-T (IL-15 armored)	Liver cancer; RMS; malignant rhabdoid tumor; liposarcoma; Wilms tumor; yolk sac tumor	GPC3-directed CAR-T cells expressing IL-15 to enhance persistence and tumor infiltration; detailed IL-15 construct design not disclosed	NCT04377932
2021	GPC3 CAR-T (IL-15 armored)	Liver cancer; RMS; malignant rhabdoid tumor; liposarcoma; tumor; Wilms tumor; yolk sac tumor	GPC3 CAR-T cells with IL-15 expression module designed to enhance immune activation and antitumor responses; IL-15 construct details not reported	NCT05103631
2022	CD70 CAR-T (IL-15 armored)	B-cell lymphoma; MDS; AML	CD70-targeting CAR-T cells engineered to express IL-15 to enhance immune expansion and persistence; IL-15 expression system details not disclosed	NCT05092451
2022	PD-L1 armored T cells (IL-15 engineered)	Advanced/metastatic/refractory NSCLC	Engineered T cells targeting PD-L1 with IL15 expression module to enhance immune activation and persistence; IL-15 construct details not publicly available	NCT05334329
2024	CD5 CAR-T (IL-15 armored)	T-cell malignancies; mantle cell lymphoma; CLL	CD5-directed CAR-T cells engineered with IL-15 expression to support proliferation and antitumor activity; IL-15 promoter and variant not disclosed	NCT05110742
2024	GPC3 CAR-T (IL-15 armored)	Liver cancer; RMS; malignant rhabdoid tumor; liposarcoma; Wilms tumor; yolk sac tumor	GPC3-targeted CAR-T cells expressing IL-15 to enhance persistence and tumor killing; IL-15 construct details not publicly disclosed	NCT04715191

The table summarizes the engineered cell type, targeted tumor-associated antigen (TAA), and available information regarding IL-15 expression constructs where reported. Abbreviations: IL-15, interleukin-15; DLBCL, diffuse large B-cell lymphoma; FL, follicular lymphoma; MCL, mantle cell lymphoma; ALL, acute lymphoblastic leukemia; CLL, chronic lymphocytic leukemia; NHL, non-Hodgkin lymphoma; RMS, rhabdomyosarcoma; MDS, myelodysplastic syndrome; AML, acute myeloid leukemia; NSCLC, non-small cell lung cancer.

survival, and antitumor activity in preclinical models while retaining a pharmacologic safety mechanism to eliminate the cells in the event of toxicity (73). Building on this concept, Hurton et al. developed CAR-T cells expressing a membrane-bound IL-15 (mbIL-15) fusion protein, enabling sustained autocrine and juxtacrine IL-15 signaling that promoted long-term persistence and memory-like phenotypes without excessive systemic cytokine release (82). Similar strategies were subsequently applied to solid tumor targets, including IL13R α -CAR (83) and Fn14-CAR (84) constructs for glioblastoma, where IL-15 armoring enhanced intratumoral persistence and antitumor efficacy in otherwise highly immunosuppressive environments.

Encouraged by these preclinical findings, IL-15-armed CAR-T cells have progressed into early clinical evaluation. In 2021, IL-15-

expressing CAR-T cells were administered to patients with relapsed or refractory B-cell acute lymphoblastic leukemia (B-ALL) following failure of both CD19- and CD22-directed CAR-T therapies Sun et al. (85), demonstrating feasibility and clinical activity in a heavily pretreated population. A phase I study published the same year further supported the safety of IL-15 incorporation, showing enhanced CAR-T expansion and persistence with manageable toxicity profiles (85). Collectively, these early clinical experiences suggested that incorporating IL-15 and its receptor complex represents a feasible and potentially effective strategy to potentiate CAR-T cell therapy.

More recent preclinical work has extended IL-15 armoring strategies to the tumor stroma. FAP-targeted CAR-T cells engineered to secrete IL-15 demonstrated robust antitumor

activity in solid tumor models by enhancing CAR-T survival, promoting memory-like differentiation, and reshaping the tumor microenvironment, highlighting the versatility of IL-15-based CAR-T engineering beyond direct tumor cell targeting (86). Importantly, definitive clinical and translational evidence for the utility of IL-15-armed CAR-T cells in solid tumors was recently provided by a study reporting outcomes from patients treated with GPC3-targeted CAR-T cells co-expressing IL-15 (71). Compared with conventional GPC3 CAR-T cells, IL-15-armed CAR-T cells exhibited markedly enhanced *in vivo* expansion, persistence, and tumor infiltration, translating into improved disease control in a subset of patients with advanced solid tumors. Although IL-15 co-expression increased the incidence of cytokine-related toxicities, these adverse events were manageable using standard interventions and activation of the inducible caspase-9 safety switch. Notably, integrated correlative analyses revealed distinct transcriptional programs in tumor-infiltrating CAR-T cells associated with clinical response, providing mechanistic insight into how IL-15 enhances CAR-T functional fitness within the hostile solid tumor microenvironment.

Collectively, these studies establish IL-15 armoring as a powerful strategy to enhance CAR-T cell persistence and efficacy across both hematologic malignancies and solid tumors, while underscoring the importance of incorporating safety mechanisms to balance potency and tolerability.

4.4.2 IL-15-armed CAR-NK cell therapy

NK cells represent an attractive platform for adoptive cell therapy due to their intrinsic cytotoxicity, lack of graft-versus-host disease risk, and suitability for allogeneic “off-the-shelf” use (87). However, the clinical application of CAR-NK cells has historically been limited by insufficient *in vivo* persistence and expansion (74). Given the central role of IL-15 in NK cells development, survival, and activation, genetic incorporation of IL-15 into CAR-NK cells has emerged as a rational strategy to enhance their therapeutic potency. Preclinical studies demonstrated that CAR-NK cells engineered to express IL-15 or IL-15-based receptor complexes exhibit markedly improved proliferation, metabolic fitness, and resistance to apoptosis, leading to enhanced antitumor activity across hematologic and solid tumor models.

In contrast to systemic cytokine administration, autocrine or membrane-associated IL-15 delivery enables sustained NK-cell support while minimizing systemic exposure and toxicity. These findings were translated into the clinic with the development of allogeneic CD19-specific CAR-NK cells derived from umbilical cord blood and engineered to express IL-15. Initial first-in-human results reported in 2020 established the feasibility and safety of this approach (74), demonstrating rapid tumor responses in patients with relapsed or refractory CD19-positive lymphoid malignancies without cytokine release syndrome, neurotoxicity, or graft-versus-host disease. Importantly, IL-15 expression supported robust *in vivo* expansion and persistence of CAR-NK cells for several weeks to months, exceeding the typically short persistence of unmodified NK-cell therapies.

Subsequent long-term follow-up and expanded cohort analyses from the same phase 1/2 clinical trial provided a more comprehensive assessment of efficacy, durability, and biological determinants of response. Mature data published in 2024 confirmed durable clinical remissions in a subset of patients and identified immune correlates associated with treatment response, including CAR-NK cell persistence, activation status, and transcriptional programs linked to cytotoxic function Marin et al. (88). These analyses demonstrated that IL-15 armoring not only enhances early expansion but also sustains functional competence of CAR-NK cells over time, addressing a central limitation of earlier NK-cell-based therapies. Beyond hematologic malignancies, IL-15-armed CAR-NK strategies are being actively explored for solid tumors. Preclinical studies have shown that IL-15 expression improves CAR-NK infiltration, survival, and antitumor efficacy within immunosuppressive tumor microenvironments, supporting broader application of this platform (89). Collectively, these findings position IL-15-armed CAR-NK cells as a promising next-generation cellular immunotherapy that combines the safety advantages of NK cells with enhanced persistence and potency conferred by IL-15 signaling.

4.4.3 IL-15 armed unconventional CAR T cells

Beyond conventional $\alpha\beta$ T cells and NK cells, IL-15 has also been incorporated into adoptive immunotherapies based on unconventional T-cell subsets, including $\gamma\delta$ T cells and invariant natural killer T (iNKT) cells. Unlike classical T cells that rely on major histocompatibility complex (MHC) restricted antigen recognition, these unconventional T cells recognize non-polymorphic ligands, thereby minimizing the risk of graft-versus-host disease (GvHD) and supporting their use as allogeneic, off-the-shelf cellular therapies.

Early preclinical work demonstrated the feasibility of this strategy in $\gamma\delta$ T cells. Makkouk et al. engineered $\gamma\delta$ T cells with a glypican-3 (GPC3)-specific CAR incorporating a 4-1BB/CD3 ζ signaling domain together with IL-15 (90). In xenograft models of hepatocellular carcinoma, IL-15-armed CAR $\gamma\delta$ T cells exhibited superior proliferation, persistence, and antitumor activity compared with CAR $\gamma\delta$ T cells lacking IL-15, underscoring the ability of IL-15 to enhance functional fitness in this unconventional T-cell platform. Similar IL-15-based engineering approaches have been extended to iNKT cells, which combine innate-like cytotoxicity with favorable tumor-homing properties. Xu and colleagues developed a GD2-targeted CAR optimized for iNKT cells that co-expresses IL-15, enabling enhanced expansion and survival of the engineered cells (76). This strategy advanced into early-phase clinical testing in children with relapsed or refractory neuroblastoma (ClinicalTrials.gov identifier: NCT03294954) (76). Initial clinical observations indicated that IL-15-armed GD2-CAR iNKT cells expanded *in vivo*, trafficked to tumor sites, and were well tolerated, supporting the feasibility of this approach.

Recent clinical advances have highlighted the therapeutic potential of invariant natural killer T (iNKT) cells engineered with chimeric antigen receptors (CARs) for the treatment of

high-risk neuroblastoma. In the first-in-human phase 1 study evaluating autologous GD2-specific CAR iNKT cells co-expressing interleukin-15 (GD2-CAR.15), treatment was well tolerated, with no dose-limiting toxicities and only a single, manageable episode of grade 2 cytokine release syndrome (91). Collectively, these emerging clinical and translational findings reinforce the concept that IL-15 armoring can substantially improve the efficacy of $\gamma\delta$ T- and iNKT-cell-based therapies, broadening the scope of IL-15-enabled adoptive cell platforms beyond conventional CAR-T and CAR-NK approaches.

4.5 IL15-based combination immunotherapy strategies

Several interventions combining IL-15 agents with checkpoint inhibitors, antibodies stimulating immune cell functions (e.g., anti-CD40, anti-CD16), monoclonal antibodies specific for tumor-associated antigens, and immunotoxins have been tested in preclinical models, with promising results.

4.5.1 Combination of IL-15 with checkpoint inhibitors

IL-15-mediated immune activation can also induce compensatory immunoregulatory mechanisms that may limit sustained antitumor responses. For instance, IL-15 stimulation has been associated with the upregulation of inhibitory receptors such as T-cell immunoglobulin and mucin domain-3 (TIM-3) and T-cell immunoreceptor with Ig and ITIM domains (TIGIT) on CD8⁺ T cells, accompanied by increased secretion of the anti-inflammatory cytokine IL-10, which may attenuate effector immune activity within the tumor microenvironment (92, 93). These observations provide a strong rationale for combining IL-15-based therapies with immune checkpoint blockade in order to counteract adaptive immune resistance and sustain cytotoxic lymphocyte function.

Preclinical studies have demonstrated encouraging outcomes for such combination approaches. In murine tumor models, subcutaneous administration of rhIL-15 together with antibodies targeting CTLA-4 and PD-L1 was evaluated in CT26 and MC38 colon carcinoma models as well as in the Transgenic Adenocarcinoma of Mouse Prostate C2 (TRAMP-C2) model. Although treatment with each individual agent produced only limited therapeutic activity, the triple combination therapy resulted in a marked enhancement of antitumor efficacy, highlighting the synergistic potential of integrating IL-15-driven immune activation with checkpoint inhibition (92).

Similar findings have been reported with IL-15 superagonists. The IL-15/IL-15R α superagonist complex N-803 has shown significant synergy when combined with checkpoint inhibitors, promoting the development of durable immune responses and further enhancing the antitumor effects of the individual treatments (94, 95). Likewise, the IL-15 receptor ligand fusion protein RLI has been demonstrated to potentiate the therapeutic efficacy of PD-1 blockade in preclinical tumor models (96). In

addition to the well-established PD-1/PD-L1 and CTLA-4 pathways, emerging inhibitory receptors are also being investigated as potential combinatorial targets. Among these, TIGIT has recently gained attention as a promising checkpoint molecule for integration with IL-15-based immunotherapeutic strategies (97).

4.5.2 Combination of IL-15 with antibodies

In addition to checkpoint blockade, IL-15-based therapies have also been investigated in combination with agents that enhance antigen presentation and immune cell activation. One example involves the use of agonistic anti-CD40 antibodies, which can restore CD4⁺ T-cell helper activity and thereby promote the generation of tumor-specific CD8⁺ T cells. In the TRAMP-C2 prostate cancer model, co-administration of IL-15 with anti-CD40 antibodies resulted in a substantial expansion of TRAMP-C2-specific CD8⁺ T cells approximately a ten-fold increase, leading to improved tumor growth control compared with either treatment alone (98–100). These findings highlight the potential of combining IL-15-driven lymphocyte activation with strategies that enhance T-cell priming and help.

Another major application of IL-15 combinations exploits its potent stimulatory effects on NK cells. Because NK cells mediate antibody-dependent cellular cytotoxicity (ADCC), IL-15 has been extensively evaluated alongside therapeutic monoclonal antibodies to amplify antitumor responses. Preclinical studies demonstrated that recombinant human IL-15 (sch rhIL-15) enhances the efficacy of several clinically relevant antibodies, including cetuximab in breast cancer models (101, 102), rituximab in chronic lymphocytic leukemia and CD20-expressing lymphoma models (103, 104), and alemtuzumab in xenograft models of adult T-cell leukemia/lymphoma (103). These combinations improve NK-cell activation and significantly increase ADCC-mediated tumor cell killing.

IL-15 superagonists have also been integrated into antibody-based immunotherapy strategies. The IL-15 superagonist complex N-803 has shown strong synergy with anti-CD20 monoclonal antibodies by enhancing NK-cell activation and ADCC against B-cell lymphomas (105, 106). Building on this concept, engineered antibody platforms such as bispecific and trispecific killer engagers (BiKEs and TriKEs) have been developed to further potentiate NK-cell responses. These molecules form an immunological bridge between NK cells and tumor cells by incorporating single-chain variable fragments (scFvs) targeting the CD16 activating receptor on NK cells and tumor-associated antigens such as CD19, CD20, or CD33, while simultaneously delivering IL-15 to support NK-cell survival and expansion *in vivo* (107–109).

More recently, fusion proteins incorporating IL-15 agonists have been designed to enhance both NK- and T-cell-mediated antitumor activity. For example, a fusion construct combining the IL-15 receptor ligand (RLI) with an anti-GD2 antibody targeting the tumor-associated disialoganglioside GD2 demonstrated potent antitumor activity in EL4 and metastatic NXS2 mouse models, primarily through enhanced ADCC (110). In addition,

multifunctional fusion proteins integrating tumor-specific antibodies, RLI, and co-stimulatory ligands such as 4-1BBL, OX40L, or glucocorticoid-induced TNF receptor ligand (GITRL) have shown the ability to reduce lung metastasis in the B16-FAP melanoma model. These constructs promoted T-cell proliferation, increased cytotoxic activity, and stimulated interferon- production, further illustrating the versatility of IL-15-based combination platforms in cancer immunotherapy (111, 112).

4.5.2 Combination of IL-15 with γ -chain family of cytokines

As mentioned before, members of the common γ -chain cytokine family, including IL-2, IL-7, IL-15, and IL-21, share receptor signaling pathways and collectively regulate lymphocyte proliferation, survival, and differentiation. Increasing evidence suggests that combining IL-15 with other γ -chain cytokines may enhance immune activation and antitumor responses. For example, IL-7 and IL-15 have complementary roles in maintaining lymphocyte homeostasis and promoting immune recovery (113). In addition, γ -chain cytokines cooperate in regulating peripheral T-cell expansion and survival (114). Experimental studies further indicate that dual cytokine strategies can enhance antitumor immunity. For instance, stimulation of lymphocytes from breast cancer patients with IL-7 and IL-15 significantly increased lymphocyte proliferation and viability compared with single cytokine treatment (115). Similarly, a recombinant IL-7/IL-15 fusion cytokine demonstrated enhanced antitumor activity in murine melanoma and colon cancer models, which was associated with increased infiltration of T cells, dendritic cells, and NK cells, as well as reduced regulatory T-cell populations (116). These findings suggest that combining IL-15 with other γ -chain cytokines may represent a promising strategy to improve antitumor immune responses.

5 Future directions

IL-15 has emerged as a cornerstone cytokine for next-generation cancer immunotherapy because of its unique capacity to expand and sustain NK cells and CD8⁺ T cells without promoting regulatory T-cell dominance or activation-induced cell death. Accumulating preclinical and clinical evidence supports the therapeutic potential of IL-15 to drive durable antitumor immunity (93). However, IL-15 signaling is embedded within a highly interconnected cytokine network, and increasing evidence suggests that certain IL-15 isoforms or receptor complexes may also contribute to immune dysregulation and tumor progression under pathological conditions (117). Elevated levels of soluble IL-15 or IL-15R α in cancer patients correlate with adverse outcomes, likely reflecting chronic inflammatory signaling rather than a direct oncogenic effect (118). These findings highlight the need to define

context-dependent IL-15 signaling states using genetic and systems-level approaches.

One major translational challenge is controlling the intensity and distribution of IL-15 signaling. Although IL-15 superagonists such as N-803 exhibit enhanced stability and biological activity, their extended half-life requires high clinical doses that can result in dose-limiting toxicities (119). Ongoing clinical trials of IL-15 derivatives will be critical for identifying therapeutic windows that maximize immune activation while limiting systemic inflammation. In addition to dosing and pharmacokinetic constraints, IL-15-driven immune activation is also regulated by inhibitory checkpoint pathways within the tumor microenvironment. Emerging evidence suggests that IL-15 stimulation can induce the upregulation of immune checkpoint molecules such as PD-1 on activated CD8⁺ T cells and PD-L1 on tumor or myeloid cells. This feedback mechanism may function as a physiological brake that prevents excessive immune activation but may also limit the efficacy of IL-15 monotherapy. Consistent with this concept, previous studies have reported increased PD-1/PD-L1 signaling following IL-15-mediated immune activation (120). These findings provide a strong mechanistic rationale for combining IL-15-based therapies with immune checkpoint blockade, an approach that has already informed the development of several next-generation immunocytokines and combination strategies in cancer immunotherapy.

IL-15-armed adoptive cell therapies represent one of the most promising applications of IL-15 biology. CAR-T, CAR-NK, CAR-NKT, and $\gamma\delta$ CAR-T cells engineered to express IL-15 or IL-15/IL-15R α complexes demonstrate enhanced persistence and superior antitumor activity across hematologic and solid malignancies. Importantly, most studies have not observed autonomous growth or malignant transformation of IL-15-engineered cells, reflecting the localized, low-level production of IL-15 within the tumor microenvironment. Nevertheless, excessive IL-15-driven immune expansion can be hazardous, as illustrated by lethal toxicity in certain AML xenograft models and the transient myelotoxicity observed in CAR-NK clinical trials (74). These data emphasize the need for inducible or spatially restricted IL-15 expression strategies in future cellular therapies.

Another critical consideration is the temporal dynamics of IL-15 exposure. While sustained IL-15 signaling supports CD8⁺ T-cell memory and effector function, chronic IL-15 stimulation of NK cells can induce functional hyporesponsiveness and alter receptor balance (121). Intermittent cytokine delivery (122) or combinatorial priming with IL-12, IL-15, and IL-21 to generate memory-like NK cells (123, 124) may provide strategies to preserve NK-cell potency while avoiding exhaustion. Collectively, future IL-15-based therapies will require precision control over cytokine dose, timing, and localization. Advances in cytokine engineering, immune-cell programming, and biomarker-guided patient selection are expected to enable safer and more effective deployment of IL-15, positioning it as a central component of next-generation cancer immunotherapy.

6 Conclusion

In summary, the development of IL-15 as an immunotherapeutic agent has introduced new opportunities beyond traditional cytokine strategies. While further work is needed to refine its safety profile and therapeutic performance in cellular platforms, ongoing innovation is anticipated to fuel meaningful advances in IL-15-based immunotherapies over the coming decades.

Author contributions

HG: Methodology, Writing – review & editing, Writing – original draft, Visualization. TM: Visualization, Writing – review & editing, Writing – original draft. QJ: Visualization, Writing – review & editing, Writing – original draft. LG: Writing – original draft, Visualization, Writing – review & editing. JL: Visualization, Writing – review & editing, Writing – original draft. SW: Writing – review & editing, Visualization, Writing – original draft. ZL: Conceptualization, Writing – original draft, Supervision, Writing – review & editing. ZZ: Writing – review & editing, Conceptualization, Supervision, Writing – original draft. GW: Writing – original draft, Conceptualization, Supervision, Writing – review & editing. WH: Conceptualization, Writing – review & editing, Supervision, Writing – original draft. FZ: Conceptualization, Supervision, Writing – review & editing, Writing – original draft.

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References

- Wang S, Yao L, Cai J, Gao H, Zhou F. Immunocytokines in cancer immunotherapy: opportunities and challenges. *Front Oncol.* (2026) 15:1716612. doi: 10.3389/fonc.2025.1716612. PMID:
- Cortés-Selva D, Dasgupta B, Singh S, Grewal IS. Innate and innate-like cells: the future of chimeric antigen receptor (CAR) cell therapy. *Trends Pharmacol Sci.* (2021) 42:45–59. doi: 10.1016/j.tips.2020.11.004. PMID:
- Kraehenbuehl L, Weng C-H, Eghbali S, Wolchok JD, Merghoub T. Enhancing immunotherapy in cancer by targeting emerging immunomodulatory pathways. *Nat Rev Clin Oncol.* (2022) 19:37–50. doi: 10.1038/s41571-021-00552-7. PMID:
- Propper DJ, Balkwill FR. Harnessing cytokines and chemokines for cancer therapy. *Nat Rev Clin Oncol.* (2022) 19:237–53. doi: 10.1038/s41571-021-00588-9. PMID:
- Floros T, Tarhini AA. Anticancer cytokines: biology and clinical effects of interferon- α 2, interleukin (il)-2, il-15, il-21, and il-12. *Semin Oncol.* (2015) 42:539–48. doi: 10.1053/j.seminoncol.2015.05.015. PMID:
- Chulpanova DS, Kitaeva KV, Green AR, Rizvanov AA, Solovyeva VV. Molecular aspects and future perspectives of cytokine-based anti-cancer immunotherapy. *Front Cell Dev Biol.* (2020) 8:402. doi: 10.3389/fcell.2020.00402. PMID:
- Yang Y, Lundqvist A. Immunomodulatory effects of il-2 and il-15; implications for cancer immunotherapy. *Cancers.* (2020) 12:3586. doi: 10.3390/cancers12123586. PMID:
- Lee S, Margolin K. Cytokines in cancer immunotherapy. *Cancers.* (2011) 3:3856–93. doi: 10.3390/cancers3043856. PMID:
- Waldmann TA. The biology of interleukin-2 and interleukin-15: implications for cancer therapy and vaccine design. *Nat Rev Immunol.* (2006) 6:595–601. doi: 10.1038/nri1901. PMID:
- Mac'Cheever MA. Twelve immunotherapy drugs that could cure cancers. *Immunol Rev.* (2008) 222:357–68. doi: 10.1111/j.1600-065x.2008.00604.x. PMID:
- Berraondo P, Sanmamed MF, Ochoa MC, Etzeberria I, Aznar MA, Pérez-Gracia JL, et al. Cytokines in clinical cancer immunotherapy. *Br J Cancer.* (2019) 120:6–15. doi: 10.1038/s41416-018-0328-y. PMID:
- Sadlack B, Merz H, Schorle H, Schimpl A, Feller AC, Horak I. Ulcerative colitis-like disease in mice with a disrupted interleukin-2 gene. *Cell.* (1993) 75:253–61. doi: 10.1016/0092-8674(93)80067-o. PMID:
- Malek TR, Porter BO, Codias EK, Scibelli P, Yu A. Normal lymphoid homeostasis and lack of lethal autoimmunity in mice containing mature t cells with severely impaired il-2 receptors. *J Immunol.* (2000) 164:2905–14. doi: 10.4049/jimmunol.164.6.2905. PMID:
- Malek TR, Yu A, Vincek V, Scibelli P, Kong L. Cd4 regulatory t cells prevent lethal autoimmunity in il-2 β -deficient mice: implications for the nonredundant function of il-2. *Immunity.* (2002) 17:167–78. doi: 10.1016/s1074-7613(02)00367-9. PMID:

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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15. Spolski R, Li P, Leonard WJ. Biology and regulation of il-2: from molecular mechanisms to human therapy. *Nat Rev Immunol.* (2018) 18:648–59. doi: 10.1038/s41577-018-0046-y. PMID:
16. Kennedy MK, Glaccum M, Brown SN, Butz EA, Viney JL, Embers M, et al. Reversible defects in natural killer and memory cd8 t cell lineages in interleukin 15-deficient mice. *J Exp Med.* (2000) 191:771–80. doi: 10.1084/jem.191.5.771. PMID:
17. Imai K, Matsuyama S, Miyake S, Suga K, Nakachi K. Natural cytotoxic activity of peripheral-blood lymphocytes and cancer incidence: an 11-year follow-up study of a general population. *Lancet.* (2000) 356:1795–9. doi: 10.1016/s0140-6736(00)03231-1. PMID:
18. Burton JD, Bamford RN, Peters C, Grant AJ, Kurys G, Goldman CK, et al. A lymphokine, provisionally designated interleukin t and produced by a human adult t-cell leukemia line, stimulates t-cell proliferation and the induction of lymphokine-activated killer cells. *Proc Natl Acad Sci.* (1994) 91:4935–9. doi: 10.1073/pnas.91.11.4935. PMID:
19. Bamford RN, Grant AJ, Burton JD, Peters C, Kurys G, Goldman CK, et al. The interleukin (il) 2 receptor beta chain is shared by il-2 and a cytokine, provisionally designated il-t, that stimulates t-cell proliferation and the induction of lymphokine-activated killer cells. *Proc Natl Acad Sci.* (1994) 91:4940–4. doi: 10.1073/pnas.91.11.4940. PMID:
20. Grabstein KH, Eisenman J, Shanebeck K, Rauch C, Srinivasan S, Fung V, et al. Cloning of a t cell growth factor that interacts with the β chain of the interleukin-2 receptor. *Science.* (1994) 264:965–8. doi: 10.1126/science.8178155. PMID:
21. Anderson DM, Johnson L, Glaccum MB, Copeland NG, Gilbert DJ, Jenkins NA, et al. Chromosomal assignment and genomic structure of il15. *Genomics.* (1995) 25:701–6. doi: 10.1016/0888-7543(95)80013-c. PMID:
22. Carson WE, Giri JG, Lindemann M, Linett ML, Ahdieh M, Paxton R, et al. Interleukin (il) 15 is a novel cytokine that activates human natural killer cells via components of the il-2 receptor. *J Exp Med.* (1994) 180:1395–403. doi: 10.1084/jem.180.4.1395. PMID:
23. Castillo EF, Schluns KS. Regulating the immune system via il-15 transpresentation. *Cytokine.* (2012) 59:479–90. doi: 10.1016/j.cyt.2012.06.017. PMID:
24. Dubois S, Mariner J, Waldmann TA, Tagaya Y. Il-15 α recycles and presents il-15 in trans to neighboring cells. *Immunity.* (2002) 17:537–47. doi: 10.1016/s1074-7613(02)00429-6. PMID:
25. Hutchins NA, Unsinger J, Hotchkiss RS, Ayala A. The new normal: immunomodulatory agents against sepsis immune suppression. *Trends Mol Med.* (2014) 20:224–33. doi: 10.1016/j.molmed.2014.01.002. PMID:
26. Fehniger TA, Caligiuri MA. Interleukin 15: biology and relevance to human disease. *Blood J Am Soc Hematol.* (2001) 97:14–32. doi: 10.1182/blood.v97.1.14. PMID:
27. Ma A, Koka R, Burkett P. Diverse functions of il-2, il-15, and il-7 in lymphoid homeostasis. *Annu Rev Immunol.* (2006) 24:657–79. doi: 10.1146/annurev.immunol.24.021605.090727. PMID:
28. Mattei F, Schiavoni G, Belardelli F, Tough DF. Il-15 is expressed by dendritic cells in response to type i ifn, double-stranded rna, or lipopolysaccharide and promotes dendritic cell activation. *J Immunol.* (2001) 167:1179–87. doi: 10.4049/jimmunol.167.3.1179. PMID:
29. Ahmad R, Ennaci J, Cordeiro P, El Bassam S, Menezes J. Herpes simplex virus-1 up-regulates il-15 gene expression in monocyctic cells through the activation of protein tyrosine kinase and pkc zeta/lambda signaling pathways. *J Mol Biol.* (2007) 367:25–35. doi: 10.1016/j.jmb.2006.12.060. PMID:
30. Lucas M, Schachterle W, Oberle K, Aichele P, Diefenbach A. Dendritic cells prime natural killer cells by trans-presenting interleukin 15. *Immunity.* (2007) 26:503–17. doi: 10.1016/j.immuni.2007.03.006. PMID:
31. Lodolce JP, Boone DL, Chai S, Swain RE, Dassopoulos T, Tretton S, et al. Il-15 receptor maintains lymphoid homeostasis by supporting lymphocyte homing and proliferation. *Immunity.* (1998) 9:669–76. doi: 10.1016/s1074-7613(00)80664-0. PMID:
32. Damjanovich S, Bene L, Matkó J, Alileche A, Goldman CK, Sharrow S, et al. Preassembly of interleukin 2 (il-2) receptor subunits on resting kit 225 k6 t cells and their modulation by il-2, il-7, and il-15: A fluorescence resonance energy transfer study. *Proc Natl Acad Sci.* (1997) 94:13134–9. doi: 10.1073/pnas.94.24.13134. PMID:
33. Dubois S, Shou W, Haneline LS, Fleischer S, Waldmann TA, Müller JR. Distinct pathways involving the flk506-binding proteins 12 and 12.6 underlie il-2-versus il-15-mediated proliferation of t cells. *Proc Natl Acad Sci.* (2003) 100:14169–74. doi: 10.1073/pnas.2335979100. PMID:
34. Schluns KS, Klonowski KD, Lefranc,ois L. Transregulation of memory cd8 t-cell proliferation by il-15+ bone marrow-derived cells. *Blood.* (2004) 103:988–94. doi: 10.1182/blood-2003-08-2814. PMID:
35. Chirifu M, Hayashi C, Nakamura T, Toma S, Shuto T, Kai H, et al. Crystal structure of the il-15–il-15 α complex, a cytokine-receptor unit presented in trans. *Nat Immunol.* (2007) 8:1001–7. doi: 10.1038/ni1492. PMID:
36. U.S. Food and Drug Administration. *FDA approves nogapendekin alfa inbakicept-pmln for BCG-unresponsive non-muscle-invasive bladder cancer.* Silver Spring, MD: U.S. Food and Drug Administration (2024).
37. Gierschek F, Schlueter J, Kühnel I, Feigl FF, Schmiedel D, Prüfer M, et al. Empowering natural killer cells to combat acute myeloid leukemia: Perspective on car-nk cell therapy. *Transfusion Med Hemotherapy.* (2025) 52:42–60. doi: 10.1159/000540962. PMID:
38. Ahmed S, DiPersio JF, Essell J, Diefenbach C, Perales M-A, Llorente CC, et al. Nktr-255 vs placebo to enhance complete responses and durability following cd19-directed car-t therapy in patients with relapsed/refractory (r/r) large b-cell lymphoma (lbcl). *Transplant Cell Ther.* (2025) 31:S11–2. doi: 10.1016/j.jctc.2025.01.021. PMID:
39. Conlon K, Watson DC, Waldmann TA, Valentin A, Bergamaschi C, Felber BK, et al. Phase i study of single agent niz985, a recombinant heterodimeric il-15 agonist, in adult patients with metastatic or unresectable solid tumors. *J Immunother Cancer.* (2021) 9:e003388. doi: 10.1136/jitc-2021-003388. PMID:
40. Matuskova H, Marasek P, Mazhara V, Simonova E, Kosinova L, Danek P, et al. Novel pd-1-targeted, activity-optimized il-15 mutein sot201 acting in cis provides antitumor activity superior to pd1-il2v. *J Immunother Cancer.* (2025) 13:e010736. doi: 10.1136/jitc-2024-010736. PMID:
41. Naing A, McKean M, Rosen I, Sommerhalder D, Shaik N, Wang I-M, et al. First-in-human phase i study to evaluate safety, tolerability, pharmacokinetics, pharmacodynamics, immunogenicity, and antitumor activity of pf-07209960 in patients with advanced or metastatic solid tumors. *ESMO Open.* (2025) 10:104291. doi: 10.1016/j.esmoop.2025.104291. PMID:
42. Vyas VV, Esposito D, Sumpter TL, Broadt TL, Hartley J, Knapp GC, et al. Clinical manufacturing of recombinant human interleukin 15. i. production cell line development and protein expression in e. coli with stop codon optimization. *Biotechnol Progr.* (2012) 28:497–507. doi: 10.1002/btpr.746. PMID:
43. Tang F, Zhao L, Jiang Y, Ba D, Cui L, He W. Activity of recombinant human interleukin-15 against tumor recurrence and metastasis in mice. *Cell Mol Immunol.* (2008) 5:189–96. doi: 10.1038/cmi.2008.23. PMID:
44. Zhou Y, Husman T, Cen X, Tsao T, Brown J, Bajpai A, et al. Interleukin 15 in cell-based cancer immunotherapy. *Int J Mol Sci.* (2022) 23:7311. doi: 10.3390/ijms23137311. PMID:
45. Sneller MC, Kopp WC, Engelke KJ, Yovandich JL, Creekmore SP, Waldmann TA, et al. Il-15 administered by continuous infusion to rhesus macaques induces massive expansion of cd8+ t effector memory population in peripheral blood. *Blood J Am Soc Hematol.* (2011) 118:6845–8. doi: 10.1182/blood-2011-09-377804. PMID:
46. Lugli E, Goldman CK, Perera LP, Smedley J, Pung R, Yovandich JL, et al. Transient and persistent effects of il-15 on lymphocyte homeostasis in nonhuman primates. *Blood J Am Soc Hematol.* (2010) 116:3238–48. doi: 10.1182/blood-2010-03-275438. PMID:
47. Conlon KC, Lugli E, Welles HC, Rosenberg SA, Fojo AT, Morris JC, et al. Redistribution, hyperproliferation, activation of natural killer cells and cd8 t cells, and cytokine production during first-in-human clinical trial of recombinant human interleukin-15 in patients with cancer. *J Clin Oncol.* (2015) 33:74–82. doi: 10.1200/jco.2014.57.3329. PMID:
48. Waldmann TA, Miljkovic MD, Conlon KC. Interleukin-15 (dys)regulation of lymphoid homeostasis: Implications for therapy of autoimmunity and cancer. *J Exp Med.* (2019) 217:e20191062. doi: 10.1084/jem.20191062. PMID:
49. Wong HC, Jeng EK, Rhode PR. The il-15-based superagonist alt-803 promotes the antigen-independent conversion of memory cd8+ t cells into innate-like effector cells with antitumor activity. *Oncoimmunology.* (2013) 2:e26442. doi: 10.4161/onci.26442. PMID:
50. Chamie K, Chang SS, Kramolowsky E, Gonzalgo ML, Agarwal PK, Bassett JC, et al. Il-15 superagonist nai in bcr-unresponsive non-muscle-invasive bladder cancer. *NEJM Evidence.* (2022) 2:EVID0a2200167. doi: 10.1056/evidoa2200167. PMID:
51. Kamat AM, Sylvester RJ, Böhle A, Palou J, Lamm DL, Brausi M, et al. Definitions, end points, and clinical trial designs for non-muscle-invasive bladder cancer: recommendations from the international bladder cancer group. *J Clin Oncol.* (2016) 34:1935–44. doi: 10.1200/jco.2015.64.4070. PMID:
52. Jarow JP, Lerner SP, Kluetz PG, Liu K, Sridhara R, Bajorin D, et al. Clinical trial design for the development of new therapies for nonmuscle-invasive bladder cancer: report of a food and drug administration and american urological association public workshop. *Urology.* (2014) 83:262–5. doi: 10.1016/j.urol.2013.10.030. PMID:
53. Kamat AM, Colombel M, Sundi D, Lamm D, Boehle A, Brausi M, et al. Bcr-unresponsive non-muscle-invasive bladder cancer: recommendations from the ibcg. *Nat Rev Urol.* (2017) 14:244–55. doi: 10.1038/nrurol.2017.16. PMID:
54. Jiang S, Redelman-Sidi G. Bcr in bladder cancer immunotherapy. *Cancers.* (2022) 14:3073. doi: 10.3390/cancers14133073. PMID:
55. Sfakianos JP, Kim PH, Hakimi AA, Herr HW. The effect of restaging transurethral resection on recurrence and progression rates in individual patients with nonmuscle invasive bladder cancer treated with intravesical bacillus calmette-guérin. *J Urol.* (2014) 191:341–5. doi: 10.1016/j.juro.2013.08.022. PMID:
56. Sylvester RJ, Van Der Meijden AP, Oosterlinck W, Witjes JA, Bouffouix C, Denis L, et al. Predicting recurrence and progression in individual patients with stage ta t1 bladder cancer using eortc risk tables: a combined analysis of 2596 patients from seven eortc trials. *Eur Urol.* (2006) 49:466–77. doi: 10.1016/j.eururo.2005.12.031. PMID:
57. Gomes-Giacoa E, Miyake M, Goodison S, Sriharan A, Zhang G, You L, et al. Intravesical alt-803 and bcr treatment reduces tumor burden in a carcinogen induced bladder cancer rat model; a role for cytokine production and nk cell expansion. *PLoS One.* (2014) 9:e96705. doi: 10.1371/journal.pone.0096705. PMID:
58. Furuya H, Chan OT, Pagano I, Zhu C, Kim N, Peres R, et al. Effectiveness of two different dose administration regimens of an il-15 superagonist complex (alt-803) in an orthotopic

- bladder cancer mouse model. *J Transl Med.* (2019) 17:29. doi: 10.1186/s12967-019-1778-6. PMID:
59. Chen W, Liu N, Yuan Y, Zhu M, Hu X, Hu W, et al. Alt-803 in the treatment of non-muscle-invasive bladder cancer: Preclinical and clinical evidence and translational potential. *Front Immunol.* (2022) 13:1040669. doi: 10.3389/fimmu.2022.1040669. PMID:
60. Liu B, Zhu X, Kong L, Wang M, Spanoudis C, Chaturvedi P, et al. Bifunctional tgfb- β trap/il-15 protein complex elicits potent nk cell and cd8+ t cell immunity against solid tumors. *Mol Ther.* (2021) 29:2949–62. doi: 10.1016/j.ymthe.2021.06.001
61. Carnemolla B, Borsi L, Balza E, Castellani P, Meazza R, Berndt A, et al. Enhancement of the antitumor properties of interleukin-2 by its targeted delivery to the tumor blood vessel extracellular matrix. *Blood J Am Soc Hematol.* (2002) 99:1659–65. doi: 10.1182/blood.v99.5.1659. PMID:
62. Borsi L, Balza E, Carnemolla B, Sassi F, Castellani P, Berndt A, et al. Selective targeted delivery of tnfr to tumor blood vessels. *Blood.* (2003) 102:4384–92. doi: 10.1186/1472-6750-11-104. PMID:
63. Ng S, Deng J, Chinnadurai R, Yuan S, Pennati A, Galipeau J. Stimulation of natural killer cell-mediated tumor immunity by an il15/tgfb-neutralizing fusion protein. *Cancer Res.* (2016) 76:5683–95. doi: 10.1158/0008-5472.CAN-16-0386
64. Jochems C, Tritsch SR, Knudson KM, Gameiro SR, Rumfield CS, Pellom ST, et al. The multi-functionality of n-809, a novel fusion protein encompassing anti-pd-1l and the il-15 superagonist fusion complex. *Oncoimmunology.* (2019) 8:e1532764. doi: 10.1080/2162402x.2018.1532764. PMID:
65. Liu B, Zhu X, Kong L, Wang M, Spanoudis C, Chaturvedi P, et al. Bifunctional TGF-trap/IL-15 protein complex elicits potent NK cell and CD8+ T cell immunity against solid tumors. *Mol Ther.* (2021) 29:2949–62. doi: 10.1016/j.ymthe.2021.06.001. PMID:
66. Xu Y, Carrascosa LC, Yeung YA, Chu ML-H, Yang W, Djuretic I, et al. An engineered il15 cytokine mutein fused to an anti-pd1 improves intratumoral t-cell function and antitumor immunity. *Cancer Immunol Res.* (2021) 9:1141–57. doi: 10.1158/2326-6066.cir-21-0058. PMID:
67. Martomo SA, Lu D, Polonskaya Z, Luna X, Zhang Z, Feldstein S, et al. Single-dose anti-pd-1l/il-15 fusion protein kd033 generates synergistic antitumor immunity with robust tumor-immune gene signatures and memory responses. *Mol Cancer Ther.* (2021) 20:347–56. doi: 10.1158/1535-7163.mct-20-0457. PMID:
68. Knudson KM, Hicks KC, Ozawa Y, Schlom J, Gameiro SR. Functional and mechanistic advantage of the use of a bifunctional anti-pd-1l/il-15 superagonist. *J Immunother Cancer.* (2020) 8:e000493. doi: 10.1136/jitc-2019-000493. PMID:
69. Adkins I, Antosova Z, Behalova K, Danek P, Danova K, Fabisik M, et al. 1087 sot201 is a novel cis-acting immunocytokine targeting il-15r β to reinvigorate pd-1+ tumor infiltrating lymphocytes and potentiate anti-tumor efficacy. (2023) 11(Suppl 1): A1196. doi: 10.1136/jitc-2023-SITC2023.1087
70. Labanieh L, Mackall CL. Car immune cells: design principles, resistance and the next generation. *Nature.* (2023) 614:635–48. doi: 10.1038/s41586-023-05707-3. PMID:
71. Steffin D, Ghatwai N, Montalbano A, Rath P, Courtney AN, Arnett AB, et al. Interleukin-15-armoured gpc3 car t cells for patients with solid cancers. *Nature.* (2025) 637:940–6. doi: 10.1038/s41586-024-08261-8. PMID:
72. Brentjens RJ, Latouche J-B, Santos E, Marti F, Gong MC, Lyddane C, et al. Eradication of systemic b-cell tumors by genetically targeted human t lymphocytes costimulated by cd80 and interleukin-15. *Nat Med.* (2003) 9:279–86. doi: 10.1038/nm827. PMID:
73. Hoyos V, Savoldo B, Quintarelli C, Mahendravada A, Zhang M, Vera J, et al. Engineering cd19-specific t lymphocytes with interleukin-15 and a suicide gene to enhance their anti-lymphoma/leukemia effects and safety. *Leukemia.* (2010) 24:1160–70. doi: 10.1038/leu.2010.75. PMID:
74. Liu E, Marin D, Banerjee P, Macapinlac HA, Thompson P, Basar R, et al. Use of car-transduced natural killer cells in cd19-positive lymphoid tumors. *N Engl J Med.* (2020) 382:545–53. doi: 10.1056/nejmoa1910607. PMID:
75. Van den Eynde A, Gehrcken L, Verhezen T, Lau HW, Hermans C, Lambrechts H, et al. Il-15-secreting car natural killer cells directed toward the pan-cancer target cd70 eliminate both cancer cells and cancer-associated fibroblasts. *J Hematol Oncol.* (2024) 17:8. doi: 10.1186/s13045-024-01525-w. PMID:
76. Heczey A, Courtney AN, Montalbano A, Robinson S, Liu K, Li M, et al. Anti-gd2 car-nkt cells in patients with relapsed or refractory neuroblastoma: an interim analysis. *Nat Med.* (2020) 26:1686–90. doi: 10.1038/s41591-020-1074-2. PMID:
77. Xu X, Huang W, Heczey A, Liu D, Guo L, Wood M, et al. Nkt cells coexpressing a gd2-specific chimeric antigen receptor and il15 show enhanced *in vivo* persistence and antitumor activity against neuroblastoma. *Clin Cancer Res.* (2019) 25:7126–38. doi: 10.1158/1078-0432.ccr-19-0421. PMID:
78. Li Y-R, Zhou Y, Yu J, Kim YJ, Li M, Lee D, et al. Generation of allogeneic car-nkt cells from hematopoietic stem and progenitor cells using a clinically guided culture method. *Nat Biotechnol.* (2025) 43:329–44. doi: 10.1038/s41587-024-02226-y. PMID:
79. Ramos CA, Courtney AN, Lulla PD, Hill LC, Kamble RT, Carrum G, et al. Off-the-shelf cd19-specific car-nkt cells in patients with relapsed or refractory b-cell Malignancies. *Transplant Cell Therapy Off Publ Am Soc For Transplant Cell Ther.* (2024) 30:S41–2. doi: 10.1016/j.jctc.2023.12.072. PMID:
80. Lee D, Dunn ZS, Guo W, Rosenthal CJ, Penn NE, Yu Y, et al. Unlocking the potential of allogeneic v82 t cells for ovarian cancer therapy through cd16 biomarker selection and car/il-15 engineering. *Nat Commun.* (2023) 14:6942. doi: 10.2106/jbjs.rvw.23.00141. PMID:
81. Li J, Chen P, Ma W. The next frontier in immunotherapy: potential and challenges of car-macrophages. *Exp Hematol Oncol.* (2024) 13:76. doi: 10.1186/s40164-024-00549-9. PMID:
82. Hurton LV, Singh H, Najjar AM, Switzer KC, Mi T, Maiti S, et al. Tethered il-15 augments antitumor activity and promotes a stem-cell memory subset in tumor-specific t cells. *Proc Natl Acad Sci.* (2016) 113:E7788–97. doi: 10.1073/pnas.1610544113. PMID:
83. Krenciute G, Prinzing BL, Yi Z, Wu M-F, Liu H, Dotti G, et al. Transgenic expression of il15 improves antiglioma activity of il13r α 2-car t cells but results in antigen loss variants. *Cancer Immunol Res.* (2017) 5:571–81. doi: 10.1016/s1525-0016(16)32885-4. PMID:
84. Li G, Zhang Z, Cai L, Tang X, Huang J, Yu L, et al. Fn14-targeted bite and car-t cells demonstrate potent preclinical activity against glioblastoma. *Oncoimmunology.* (2021) 10:1983306. doi: 10.1080/2162402x.2021.1983306. PMID:
85. Sun Y, Su Y, Wang Y, Liu N, Li Y, Chen J, et al. Cd19 car-t cells with membrane-bound il-15 for b-cell acute lymphoblastic leukemia after failure of cd19 and cd22 car-t cells: case report. *Front Immunol.* (2021) 12:728962. doi: 10.3389/fimmu.2021.728962. PMID:
86. Pang Y, Chen L, Sun Q, He N, Lai Z, Wang X, et al. A new strategy for car-t therapy in solid tumors: il-15-autocrine signaling augments tumor stroma depletion and promotes a tscm subset in the tme. *Cell Death amp; Dis.* (2025) 17(1):39. doi: 10.1038/s41419-025-08405-2. PMID:
87. Perfect JR. The antifungal pipeline: a reality check. *Nat Rev Drug Discov.* (2017) 16:603–16. doi: 10.1038/nrd.2017.46. PMID:
88. Marin D, Li Y, Basar R, Rafei H, Daher M, Dou J, et al. Safety, efficacy and determinants of response of allogeneic cd19-specific car-nk cells in cd19+ b cell tumors: a phase 1/2 trial. *Nat Med.* (2024) 30:772–84. doi: 10.1038/s41591-023-02785-8. PMID:
89. Baughan SL, Folsom T, Johnson M, Kreuger J, Webber BR, Moriarity BS. Correction: Perspective: Il-15 cytokine-armored nk cells as ready-to-use immunotherapy for diverse Malignancies: therapeutic potential and toxicity risks. *Front Immunol.* (2026) 17:1779800. doi: 10.3389/fimmu.2026.1779800. PMID:
90. Makkouk A, Yang XC, Barca T, Lucas A, Turkoz M, Wong JT, et al. Off-the-shelf v81 gamma delta t cells engineered with glypican-3 (gpc-3)-specific chimeric antigen receptor (car) and soluble il-15 display robust antitumor efficacy against hepatocellular carcinoma. *J Immunother Cancer.* (2021) 9:e003441. doi: 10.1136/jitc-2021-003441. PMID:
91. Heczey A, Xu X, Courtney AN, Tian G, Barragan GA, Guo L, et al. Author correction: Anti-gd2 car-nkt cells in relapsed or refractory neuroblastoma: updated phase 1 trial interim results. *Nat Med.* (2024) 30:1210–1690. doi: 10.1038/s41591-024-02799-w. PMID:
92. Yu P, Steel JC, Zhang M, Morris JC, Waitz R, Fasso M, et al. Simultaneous administration of two regulatory t-cell subsets enhanced interleukin-15 efficacy in a prostate tumor model. *Proc Natl Acad Sci.* (2012) 109:6187–92. doi: 10.1073/pnas.1203479109. PMID:
93. Yu P, Steel JC, Zhang M, Morris JC, Waldmann TA. Simultaneous blockade of multiple immune system inhibitory checkpoints enhances antitumor activity mediated by interleukin-15 in a murine metastatic colon carcinoma model. *Clin Cancer Res.* (2010) 16:6019–28. doi: 10.1158/1078-0432.ccr-10-1966. PMID:
94. Kim PS, Kwilas AR, Xu W, Alter S, Jeng EK, Wong HC, et al. Il-15 superagonist/il-15r α sushi-fc fusion complex (il-15sa/il-15r α su-fc; alt-803) markedly enhances specific subpopulations of nk and memory cd8+ t cells, and mediates potent anti-tumor activity against murine breast and colon carcinomas. *Oncotarget.* (2016) 7:16130. doi: 10.18632/oncotarget.7470
95. Mathios D, Park C-K, Marcus WD, Alter S, Rhode PR, Jeng EK, et al. Therapeutic administration of il-15 superagonist complex alt-803 leads to long-term survival and durable antitumor immune response in a murine glioblastoma model. *Int J Cancer.* (2016) 138:187–94. doi: 10.1002/ijc.29686. PMID:
96. Desbois M, Le Vu P, Coutzac C, Marcheteau E, Beal C, Terme M, et al. Il-15 trans-signaling with the superagonist rli promotes effector/memory cd8+ t cell responses and enhances antitumor activity of pd-1 antagonists. *J Immunol.* (2016) 197:168–78. doi: 10.4049/jimmunol.1600019. PMID:
97. Judge SJ, Darrow MA, Thorpe SW, Gingrich AA, O'Donnell EF, Bellini AR, et al. Analysis of tumor-infiltrating nk and t cells highlights il-15 stimulation and tigit blockade as a combination immunotherapy strategy for soft tissue sarcomas. *J Immunother Cancer.* (2020) 8:e001355. doi: 10.1136/jitc-2020-001355. PMID:
98. Zhang M, Ju W, Yao Z, Yu P, Wei B-R, Simpson RM, et al. Augmented il-15r α expression by cd40 activation is critical in synergistic cd8 t cell-mediated antitumor activity of anti-cd40 antibody with il-15 in tramp-c2 tumors in mice. *J Immunol.* (2012) 188:6156–64. doi: 10.4049/jimmunol.1102604. PMID:
99. Zhang M, Yao Z, Dubois S, Ju W, Müller JR, Waldmann TA. Interleukin-15 combined with an anti-cd40 antibody provides enhanced therapeutic efficacy for murine models of colon cancer. *Proc Natl Acad Sci.* (2009) 106:7513–8. doi: 10.1073/pnas.0902637106. PMID:

100. Bennett SR, Carbone FR, Karamalis F, Flavell RA, Miller JF, Heath WR. Help for cytotoxic-t-cell responses is mediated by cd40 signalling. *Nature*. (1998) 393:478–80. doi: 10.1038/30996. PMID:
101. Ochoa MC, Minute L, López A, Pérez-Ruiz E, Gomar C, Vasquez M, et al. Enhancement of antibody-dependent cellular cytotoxicity of cetuximab by a chimeric protein encompassing interleukin-15. *Oncimmunology*. (2018) 7:e1393597. doi: 10.1080/2162402x.2017.1393597. PMID:
102. Roberti MP, Rocca YS, Amat M, Pampena MB, Loza J, Coló F, et al. IL-2-or il-15-activated nk cells enhance cetuximab-mediated activity against triple-negative breast cancer in xenografts and in breast cancer patients. *Breast Cancer Res Treat*. (2012) 136:659–71. doi: 10.1007/s10549-012-2287-y. PMID:
103. Moga E, Cantó E, Vidal S, Juarez C, Sierra J, Briones J. Interleukin-15 enhances rituximab-dependent cytotoxicity against chronic lymphocytic leukemia cells and overcomes transforming growth factor beta-mediated immunosuppression. *Exp Hematol*. (2011) 39:1064–71. doi: 10.1016/j.exphem.2011.08.006. PMID:
104. Zhang M, Wen B, Anton OM, Yao Z, Dubois S, Ju W, et al. IL-15 enhanced antibody-dependent cellular cytotoxicity mediated by nk cells and macrophages. *Proc Natl Acad Sci*. (2018) 115:E10915–24. doi: 10.1073/pnas.1811615115. PMID:
105. Rosario M, Liu B, Kong L, Collins LI, Schneider SE, Chen X, et al. The il-15-based alt-803 complex enhances fcγriiia-triggered nk cell responses and *in vivo* clearance of b cell lymphomas. *Clin Cancer Res*. (2016) 22:596–608. doi: 10.1158/1078-0432.ccr-15-1419. PMID:
106. Liu B, Kong L, Han K, Hong H, Marcus WD, Chen X, et al. A novel fusion of alt-803 (interleukin (il)-15 superagonist) with an antibody demonstrates antigen-specific antitumor responses. *J Biol Chem*. (2016) 291:23869–81. doi: 10.1074/jbc.m116.733600. PMID:
107. Felices M, Kodal B, Hinderlie P, Kaminski MF, Cooley S, Weisdorf DJ, et al. Novel cd19-targeted trike restores nk cell function and proliferative capacity in cll. *Blood Adv*. (2019) 3:897–907. doi: 10.1182/bloodadvances.2018029371. PMID:
108. Schmohl JU, Felices M, Taras E, Miller JS, Vallera DA. Enhanced adcc and nk cell activation of an anticarcinoma bispecific antibody by genetic insertion of a modified il-15 cross-linker. *Mol Ther*. (2016) 24:1312–22. doi: 10.1038/mt.2016.88. PMID:
109. Vallera DA, Felices M, McElmurry R, McCullar V, Zhou X, Schmohl JU, et al. Il15 trispecific killer engagers (trike) make natural killer cells specific to cd33+ targets while also inducing persistence, *in vivo* expansion, and enhanced function. *Clin Cancer Res*. (2016) 22:3440–50. doi: 10.1158/1078-0432.ccr-15-2710. PMID:
110. Vincent M, Bessard A, Cochonneau D, Teppaz G, Solé V, Maillasson M, et al. Tumor targeting of the il-15 superagonist rli by an anti-gd2 antibody strongly enhances its antitumor potency. *Int J Cancer*. (2013) 133:757–65. doi: 10.1002/ijc.28059. PMID:
111. Beha N, Harder M, Ring S, Kontermann RE, Müller D. Il15-based trifunctional antibody-fusion proteins with costimulatory tnfr-superfamily ligands in the single-chain format for cancer immunotherapy. *Mol Cancer Ther*. (2019) 18:1278–88. doi: 10.1158/1535-7163.mct-18-1204. PMID:
112. Kermer V, Baum V, Hornig N, Kontermann RE, Müller D. An antibody fusion protein for cancer immunotherapy mimicking il-15 trans-presentation at the tumor site. *Mol Cancer Ther*. (2012) 11:1279–88. doi: 10.1158/1535-7163.mct-12-0019. PMID:
113. Alpdogan Ö, van den Brink MR. Il-7 and il-15: therapeutic cytokines for immunodeficiency. *Trends Immunol*. (2005) 26:56–64. doi: 10.1016/j.it.2004.11.002. PMID:
114. Kinter AL, Godbout EJ, McNally JP, Sereti I, Roby GA, O'Shea MA, et al. The common γ-chain cytokines il-2, il-7, il-15, and il-21 induce the expression of programmed death-1 and its ligands. *J Immunol*. (2008) 181:6738–46. doi: 10.4049/jimmunol.181.10.6738. PMID:
115. Timofeeva SV, Sitkovskaya AO, Zlatnik EY, Filippova SY, Shamova TV, Mezheva IV, et al. The common gamma-chain cytokines il-2, il-7 and il-15 impact on proliferation of lymphocytes *in vitro* in breast cancer patients. (2021) 39(15_suppl): e13060. doi: 10.1200/jco.2021.39.15_suppl.e13060
116. Song Y, Liu Y, Hu R, Su M, Rood D, Lai L. *In vivo* antitumor activity of a recombinant il7/il15 hybrid cytokine in mice. *Mol Cancer Ther*. (2016) 15:2413–21. doi: 10.1158/1535-7163.MCT-16-0111
117. Chen J, Petrus M, Bamford R, Shih JH, Morris JC, Janik JE, et al. Increased serum soluble il-15rα levels in t-cell large granular lymphocyte leukemia. *Blood J Am Soc Hematol*. (2012) 119:137–43. doi: 10.1002/jmv.25187. PMID:
118. Badoual C, Bouchaud G, Agueznay NEH, Mortier E, Hans S, Gey A, et al. The soluble α chain of interleukin-15 receptor: a proinflammatory molecule associated with tumor progression in head and neck cancer. *Cancer Res*. (2008) 68:3907–14. doi: 10.1158/0008-5472.can-07-6842. PMID:
119. Isvoranu G, Surcel M, Huică R-I, Munteanu AN, Privu IR, Ciotaru D, et al. Natural killer cell monitoring in cutaneous melanoma-new dynamic biomarker. *Oncol Lett*. (2019) 17:4197–206. doi: 10.3892/ol.2019.10069. PMID:
120. Knudson KM, Hicks KC, Alter S, Schlom J, Gameiro SR. Mechanisms involved in il-15 superagonist enhancement of anti-pd-1 therapy. *J Immunother Cancer*. (2019) 7:82. doi: 10.1186/s40425-019-0551-y. PMID:
121. Elpek KG, Rubinstein MP, Bellemare-Pelletier A, Goldrath AW, Turley SJ. Mature natural killer cells with phenotypic and functional alterations accumulate upon sustained stimulation with il-15/il-15rα complexes. *Proc Natl Acad Sci*. (2010) 107:21647–52. doi: 10.1073/pnas.1012128107. PMID:
122. Felices M, Lenvik AJ, McElmurry R, Chu S, Hinderlie P, Bendzick L, et al. Continuous treatment with il-15 exhausts human nk cells via a metabolic defect. *JCI Insight*. (2018) 3:e96219. doi: 10.1172/jci.insight.96219. PMID:
123. Romee R, Schneider SE, Leong JW, Chase JM, Keppel CR, Sullivan RP, et al. Cytokine activation induces human memory-like nk cells. *Blood J Am Soc Hematol*. (2012) 120:4751–60. doi: 10.1182/blood-2012-04-419283. PMID:
124. Govindaraj S, Ibegbu C, Ali SA, Babu H, Shanmugasundaram U, Villinger F, et al. Il-15/il-15ra synergies with il-12 to induce functional cd8 t cells and nk cells during chronic shiv infection. *AIDS Res Hum Retroviruses*. (2025) 41:120–3. doi: 10.1089/aid.2024.0043. PMID: